

Intern Hub

A Project Report Submitted in Partial Fulfilment of the

Requirements for the Award of the Degree of

BACHELOR OF COMPUTER APPLICATIONS

By

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Under the guidance of

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OF COMPUTER SCIENCES
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MARCH 2026



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CERTIFICATE

*This is to certify that the report titled **Intern Hub** is a bonafide record of work done by **Devana Jayachandran (230021080677)** of **Santhigiri College of Computer Sciences** in partial fulfillment of the requirements of **Sixth Semester of Bachelor of Computer Applications** during the year **2026**.*

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DECLARATION

I **Devana Jayachandran** hereby declare that the project report, titled “**Intern Hub**” is a record of original work undertaken by me for the award of the degree of Bachelor of Computer Applications. I have completed this project under the guidance of Ms. Remya R, Assistant Professor, Department of Computer Science.

I also declare that this project has not been submitted for the award of any degree. I hereby confirm the originality of the work.

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Place :

ACKNOWLEDGEMENT

A project is not complete if one fails to acknowledge all who have been instrumental in the successful completion of the project. If words were to be the symbol of undiluted feelings and token of gratitude, then let the words play the heralding role of expressing our gratitude.

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ABSTRACT

Intern Hub is an online platform designed to connect students with companies offering internship opportunities. Whether students are searching for suitable internships or companies are looking for potential interns, InternHub provides a centralized system to meet both needs efficiently. The goal of the system is to simplify the internship search and application process while ensuring authenticity and transparency.

In addition to providing internship listings, InternHub includes a company verification process managed by the admin to ensure that only genuine organizations can offer internships. Companies can register through the guest page, and after admin approval, they can log in to post internship programs, manage applications, and interact with students. This process enhances trust and reliability within the platform.

InternHub also allows students to register, search for companies, explore available internships, and view detailed information such as course duration, syllabus, and fees. Students can apply for internships directly through the platform and track their application status. Once approved by the company, students can complete the payment process securely.

Overall, InternHub aims to create a transparent, efficient, and user-friendly environment that benefits students, companies, and administrators. By integrating internship discovery, application management, approval workflows, and payment processing into a single system, InternHub serves as a comprehensive solution for internship management.

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ABBREVIATIONS

IDE	Integrated Development Environment
CPU	Central Processing Unit
DBMS	Data Base Management System
RDBMS	Relational Data Base Management System
NF	Normal Forms
PK	Primary Key
FK	Foreign Key
DFD	Data Flow Diagram

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1. INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

InternHub is an online internship management platform designed to connect students with companies offering internship opportunities through a centralized and secure digital system. In today's competitive world, internships are essential for students to gain practical knowledge, real-world experience, and career readiness. However, finding genuine online internships and managing applications can be challenging due to scattered information, lack of trusted platforms, and absence of proper verification systems.

The motivation behind InternHub is to create a reliable and transparent online environment where students can easily discover verified internship opportunities and companies can efficiently recruit suitable candidates. Many students struggle to identify authentic companies, especially in remote or online internship settings. At the same time, companies face difficulty in reaching skilled and eligible students. InternHub bridges this gap by offering a structured platform for internship discovery, online application submission, status tracking, and communication.

A key feature of InternHub is its admin approval mechanism. Companies must register and get verified by the admin before they can post internship opportunities. This ensures authenticity, prevents fraudulent activities, and maintains trust within the platform.

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1.2 THE PROPOSED SYSTEM

The proposed system for InternHub is a web-based platform that manages internship opportunities, applications, approvals, and payments in an organized manner. In this system, companies can register through a guest page. After registration, the admin verifies the company details and either approves or rejects the request. Only approved companies can log in and access the system. Once logged in, companies can add internship opportunities, edit or delete them, and view applications submitted by students. Companies can also approve or reject student applications based on their requirements.

Students can register and log in to the platform to search for companies and available internships. They can view detailed information about internships, including course duration, syllabus, and fees. Students can apply for internships of their choice, and their applications will be sent to the respective companies for approval. After approval by the company, students can proceed with payment for the internship. A portion of the payment (10%) will be allocated to the admin as a service charge, while the remaining amount goes to the company. The system also provides application tracking, payment management, and communication features to ensure a smooth internship process for all users.

1.3. PROJECT SCOPE

1.3.1. Limitations of the Existing System

- Difficulty in finding genuine internships: Students often struggle to identify trusted companies and authentic internship opportunities.
- Lack of centralized platform: Internship information is scattered across multiple websites and sources, making the search process confusing and inefficient.
- Manual application process: Students must apply separately on different company portals, leading to time-consuming procedures.
- No verification of companies: Existing systems do not ensure whether the companies offering internships are legitimate or approved.
- Poor communication: Students receive limited updates regarding application status, approval, or rejection.
- Limited transparency: There is no clear tracking system for internship applications and approvals.
- Difficulty in managing applications: Companies find it hard to handle large numbers of applications manually.
- Payment issues: Internship payments are often handled outside the system, leading to lack of security and transparency.
- Lack of monitoring: No proper mechanism exists to monitor internship processes and transactions.
- Limited accessibility: Students from remote areas face difficulties accessing internship opportunities

1.3.2. Advantages of the Proposed System

- **Centralized Internship Platform:** InternHub provides a single platform where students can search and apply for internships offered by verified companies.
- **Verified Companies:** Companies must be approved by the admin before accessing the system, ensuring authenticity and trust.
- **Easy Application Process:** Students can apply for internships online without visiting company websites separately.
- **Internship Details Access:** Students can view course duration, syllabus, and fees before applying.
- **Application Tracking:** Students can monitor the status of their applications in real time.
- **Efficient Company Management:** Companies can add, edit, and manage internship postings easily.
- **Application Approval System:** Companies can approve or reject student applications digitally.
- **Secure Payment System:** Approved students can make payments through the platform safely.
- **Admin Revenue Management:** The admin receives a commission (10%) from internship payments automatically.
- **Improved Communication:** Notifications and updates keep students, companies, and admin informed.
- **Scalable and Reliable:** The system can handle a large number of users, internships, and applications efficiently.

2. SYSTEM ANALYSIS

2.1 INTRODUCTION

Software Engineering is the analysis, design, construction, verification and management of technical or social entities. To engineer software accurately, a software engineering process must be defined. System analysis is a detailed study of the various operations performed by the system and their relationship within and module of the system. It is a structured method for solving the problems related to the development of a new system. The detailed investigation of the present system is the focal point of system analysis. This phase involves the study of parent system and identification of system objectives. Information has to be collected from all people who are affected by or who use the system. During analysis, data are collected on the variable files, decision point and transactions handled by the present system. The main aim of system is to provide the efficient and user-friendly automation. So, the system analysis process should be performed with extreme precision, so that an accurate picture of existing system, its disadvantages and the requirements of the new system can be obtained. System analysis involves gathering the necessary information and using the structured tool for analysis. This includes the studying existing system and its drawback, designing a new system and conducting cost benefit analysis. System analysis is a problemsolving activity that requires intensive communication between the system users and system developers. The system is studied to the minute detail and analysed. The system is viewed as a whole and the inputs to the system are identified. The outputs from the organization are traced through various phases of processing of inputs.

There are a number of different approaches to system analysis. When a computer-based information system is developed, systems analysis (according to the Waterfall model) would constitute the following steps:

- The development of a feasibility study, involving determining whether a project is economically, technologically and operationally feasible.
- Conducting fact-finding measures, designed to ascertain the requirements of the system's end-users. These typically span interviews, questionnaires, or visual observations of work on the existing system.

- Gauging how the end-users would operate the system (in terms of general experience in using computer hardware or software), what the system would be used for and so on. Techniques such as interviews, questionnaires etc. can be used for the detailed study of these processes. The data collected by these sources must be scrutinized to arrive at a conclusion. The conclusion is an understanding of how the system functions. This system is called the Existing System. The Existing system is then subjected to close observation and the problem areas are identified. The designer now functions as a problem solver and tries to sort out the difficulties that the enterprise faces. The solutions are given as a proposal which is the Proposed System. The proposal is then weighed with the existing system analytically and the best one is selected. The proposal is then presented to the user for an endorsement by the user. The proposal is reviewed on user request and suitable changes are made. This is a loop that ends as soon as the user is satisfied with the proposal.

2.2 STAKEHOLDERS IN THIS PROJECT

2.2.1 ADMIN

The Admin is the primary stakeholder and overall controller of the InternHub system. The admin is responsible for monitoring and managing the entire internship platform to ensure smooth operation, authenticity, and security. The admin verifies company registrations submitted through the guest page and approves or rejects them based on validity. Only approved companies are allowed to access the system, ensuring that students interact with genuine organizations. The admin also oversees student registrations, manages internship categories, and monitors all activities within the platform. Additionally, the admin supervises internship postings, application processes, and payment transactions. The system allocates 10% of the internship payment to the admin as a service charge, which the admin manages and monitors.

Furthermore, the admin generates reports related to companies, students, internships, applications, and payments. Through centralized control and continuous monitoring, the admin ensures transparency, reliability, and efficient functioning of the InternHub platform

2.2.2 Company

Companies are key stakeholders of the InternHub system as they provide internship opportunities to students. Companies can register through the guest page by submitting their details for verification. After admin approval, companies can log in to the system and access company-specific features. Once logged in, companies can add new internship programs, edit existing ones, and view internship details. They can also review applications submitted by students, examine applicant information, and approve or reject applications based on eligibility criteria. Companies are responsible for managing internship content such as course duration, syllabus, and fees. They can track approved applicants and monitor payments made by students. InternHub provides companies with an efficient platform to recruit interns, manage applications digitally, and communicate with students effectively.

2.2.3. Student

Students are the primary users of the InternHub system who utilize the platform to search and apply for internships. Students can create accounts by registering with personal and academic details and then log in to access the system. After logging in, students can search for companies, view available internships offered by selected companies, and access detailed information including duration, syllabus, and fees. Students can apply for internships of their choice, and their applications will be sent to the respective companies for approval. Once approved, students can proceed with payment for the internship through the platform. Students can also track the status of their applications, receive notifications regarding approvals or rejections, and manage their internship activities efficiently. InternHub provides students with a convenient, transparent, and reliable environment to explore verified internship opportunities and gain practical experience.

2.3 SOFTWARE REQUIREMENT SPECIFICATION

2.3.1 ADMIN

- System should have the facility to login using username and password.
- System should have the facility to view the admin homepage.
- System should have the facility to verify and approve or reject company registrations.
- System should have the facility to view and monitor all companies.
- System should have the facility to view internships posted by companies.
- System should have the facility to manage payment transactions and commission details.
- System should have the facility to generate system reports and analytics.
- System should have the facility to logout.

2.3.2 Company

- System should have the facility to login using username and password after admin approval.
- System should have the facility to view the company homepage.
- System should have the facility to add new internship programs.
- System should have the facility to edit and manage internship details.
- System should have the facility to view internships posted by the company.
- System should have the facility to view student applications.
- System should have the facility to approve or reject applications.
- System should have the facility to view approved applicants.
- System should have the facility to generate certificate for the internship
- System should have the facility to track payments made by students.
- System should have the facility to logout

2.3.3 Student

- System should have the facility to register and login using username and password.
- System should have the facility to view the student homepage.
- System should have the facility to search companies.
- System should have the facility to view internships offered by selected companies.
- System should have the facility to view internship details such as duration, syllabus, and fees.
- System should have the facility to apply for internships.
- System should have the facility to track application status.
- System should have the facility to make payment after approval.
- System should have the facility to receive notifications regarding application updates.
- System should have the facility to download the certificate of the internship.
- System should have the facility to logout

Table 2.1 Sign off table

SI.NO	NAME & DESIGNATION	DATE	ACCEPTED (YES/NO)
1	Ms.Remya R Assistant Professor Santhigiri College of Computer Sciences		
2	Devana Jayachandran Developer		

2.4 FEASIBILITY STUDY

Feasibility is defined as the practical extent to which a project can be performed successfully. To evaluate feasibility, a feasibility study is performed, which determines whether the solution considered to accomplish the requirements is practical and workable in the software. Information such as resource availability, cost estimation for software development, benefits of the software to the organization after it is developed and cost to be incurred on its maintenance are considered during the feasibility study. The objective of the feasibility study is to establish the reasons for developing the software that is acceptable to users, adaptable to change and conformable to established standards. Various other objectives of feasibility study are listed below.

- To analyse whether the software will meet organizational requirements.
- To determine whether the software can be implemented using the current technology and within the specified budget and schedule.
- To determine whether the software can be integrated with other existing software.

When our project guide as well as our client Ms.Remya R told us regarding the project and about Word to the Wise for getting the desired product developed, it comes up with rough idea about what all functions the software must perform and which all features are expected from the software.

Referencing to this information, we have done studies and discussions about whether the desired system and its functionality are feasible to develop and the output of this

phase is a feasibility study report that should contain adequate comments and recommendations. Various types of feasibility that we checked include technical feasibility, operational feasibility, and economic feasibility.

Technical Feasibility

Technical feasibility assesses the current resources (such as hardware and software) and technology, which are required to accomplish user requirements in the software within the allocated time and budget. For this, the software development team ascertains whether the current resources and technology can be upgraded or added in the software to accomplish specified user requirements. Technical feasibility also performs the following tasks.

- Analyses the technical skills and capabilities of the software development team members.
- Determines whether the relevant technology is stable and established.
- Ascertains that the technology chosen for software development has a large number of users so that they can be consulted when problems arise or improvements are required.

From our perspective there are two languages PYTHON(Django), HTML and database MySQL which are used to develop this web-based applications. PYTHON is used in the front end and MySQL is used in the back end. The Word to the Wise is web based and thus can be accessed through any browsers. As we are using these latest technologies which are currently trending and used by a number of developers across the globe, we can say that our project is technically feasible.

Operational Feasibility

Operational feasibility assesses the extent to which the required software performs a series of steps to solve business problems and user requirements. This feasibility is dependent on human resources (software development team) and involves visualizing whether the software will operate after it is developed and be operative once it is installed. Operational feasibility also performs the following tasks.

- Determines whether the problems anticipated in user requirements are of high priority.
- Determines whether the solution suggested by the software development team is acceptable.
- Analyses whether users will adapt to a new software.
- Determines whether the organization is satisfied by the alternative solutions proposed by the software development team.

We found that our project will be satisfied for the client since we were discussing every detail about the software with the client at every step. The most important part of operational feasibility study is the input from client. So, the software is built completely according to the requirements of the client. We have used the current industry standards for the software. Hence, we can say that this software is operationally feasible

Economic Feasibility

Economic feasibility determined that the required software is capable of generating financial gains for the organization. It involves the cost incurred on the software development team, estimated cost of hardware and software, cost of performing feasibility study, and so on.

For this, it is essential to consider expenses made on purchases (such as hardware purchase) and activities required to carry out software development. In addition, it is necessary to consider the benefits that can be achieved by developing the software. Software is said to be economically feasible based on following factors:-

- Cost incurred on software development to produce long-term gains for an organization.
- Cost required to conduct full software investigation (such as requirements elicitation and requirements analysis).
- Cost of hardware, software, development team, and training.

Our project is economically feasible as its development cost comes under a small budget and most of the technologies are open source and free. It's a group student project so there are no personnel costs. Development time is well planned and will not affect other operations and activities of the individuals. Once the system has been developed, the companies purchasing the system will be providing with a manual for training purposes. There is no need to purchase new hardware since the existing computers can still be used to implement the new system.

2.5 SOFTWARE DEVELOPMENT LIFECYCLE MODEL

One of the basic notions of the software development process is SDLC models which stand for Software Development Life Cycle models. SDLC – is a continuous process, which starts from the moment, when it's made a decision to launch the project, and it ends at the moment of its full remove from the exploitation. Software development lifecycle (SDLC) is a framework that defines the steps involved in the development of software. It covers the detailed plan for building, deploying and maintaining the software. SDLC defines the complete cycle of development i.e. all the tasks involved in gathering a requirement for the maintenance of a Product.

Some of the common SDLC models are Waterfall Model, V-Shaped Model, Prototype Model, Spiral Model, Iterative Incremental Model, Big Bang Model, Agile Model. We used Agile Model for our Project.

Agile Model

Agile Model is a combination of the Iterative and incremental model. This model focuses more on flexibility while developing a product rather than on the requirement. In the agile methodology after every development iteration, the client is able to see the result and understand if he is satisfied with it or he is not. Extreme programming is one of the practical uses of the agile model. The basis of this model consists of short meetings where we can review our project. In Agile, a product is broken into small

incremental builds. It is not developed as a complete product in one go. At the end of each sprint, the project guide verifies the product and after his approval, it is finalised. Client feedback is taken for improvement and his suggestions and enhancement are worked on in the next sprint. Testing is done in each sprint to minimize the risk of any failures.

Advantages of Agile Model:

- It allows more flexibility to adapt to the changes.
- The new feature can be added easily.
- Customer satisfaction as the feedback and suggestions are taken at every stage.
- Risks are minimized thanks to the flexible change process.

Disadvantages:

- Lack of documentation.
- If a customer is not clear about how exactly they want the product to be, then the project would fail.
- With all the corrections and changes there is possibility that the project will exceed expected time.

2.6 HARDWARE AND SOFTWARE REQUIREMENTS

2.6.1 SOFTWARE SPECIFICATION

This project is built upon the latest technology software.

Front end	:	HTML, JavaScript, CSS
Development tool	:	Python Django
Database	:	MySQL
Web server	:	APACHE Webserver
Operating System	:	Windows 11
Framework	:	Django, Bootstrap

2.6.1.1 PYTHON

Django, an open-source web framework written in Python, facilitates rapid and scalable web development with a clean, pragmatic design. At its core is the Model- View- Template (MVT) architectural pattern, comprising models for data structures, views for handling user requests and rendering responses, and templates for defining HTML structure. Django's Object-Relational Mapping (ORM) simplifies database interaction by translating Python models into corresponding database tables. The framework boasts a dynamic admin interface for effortless database management and customization. URL routing is declarative, providing a clear structure for web applications, while the template system allows the separation of HTML and Python code. Middleware components enhance request/response processing, enabling tasks like authentication and security checks. Django includes a form handling system for easy validation and processing of HTML forms, and its security features guard against common vulnerabilities such as Cross-Site Scripting (XSS) and Cross-Site Request Forgery (CSRF). The built-in authentication system manages user-related functionalities, while the testing framework ensures code reliability. Optionally, the Django REST framework extends capabilities for building robust Web APIs with features like serializers, authentication, and view sets. With a command-line interface (CLI) for various tasks, Django empowers developers to create scalable, secure, and maintainable

web applications efficiently, making it a preferred choice for web development projects globally.

2.6.1.2 MySQL

MySQL is the world's most popular open-source database software, with over 100 million copies of its software downloaded or distributed throughout its history. With its superior speed, reliability, and ease of use, MySQL has become the preferred choice for Web, Web 2.0, SaaS, ISV, Telecom companies and forward-thinking corporate IT Managers because it eliminates the major problems associated with downtime, maintenance and administration for modern, online applications.

Many of the world's largest and fastest-growing organizations use MySQL to save time and money powering their high-volume Web sites, critical business systems, and packaged software, including industry leaders such as Yahoo!, Alcatel-Lucent, Google, Nokia, YouTube, Wikipedia, and Booking.com.

The flagship MySQL offering is MySQL Enterprise, a comprehensive set of production- tested software, proactive monitoring tools, and premium support services available in an affordable annual subscription.

MySQL is a key part of LAMP (Linux, Apache, MySQL, PH / Perl / Python), the fast-growing open-source enterprise software stack. More and more companies are using LAMP as an alternative to expensive proprietary software stacks because of its lower cost and freedom from platform lock-in.

MySQL was originally founded and developed in Sweden by two Swedes and a Finn: David Axmark, Allan Larsson and Michael "Monty" Wideness, who had worked together since the 1980's. MySQL, the most popular Open-Source SQL database management system, is developed, distributed, and supported by Oracle Corporation.

MySQL is a database management system. A database is a structured collection of data. It may be anything from a simple shopping list to a picture gallery or the vast amounts of information in a corporate network. To add, access, and process data stored in a computer database, you need a database management system such as MySQL Server. Since computers are very good at handling large amounts of data, database management

systems play a central role in computing, as standalone utilities, or as parts of other applications.

MySQL databases are relational. A relational database stores data in separate tables rather than putting all the data in one big storeroom. The database structures are organized into physical files optimized for speed. The logical model, with objects such as databases, tables, views, rows, and columns, offers a flexible programming environment. You set up rules governing the relationships between different data fields, such as one-to-one, one-to-many, unique, required or optional, and pointers between different tables. The database enforces these rules, so that with a well-designed database, your application never sees inconsistent, duplicate, orphan, out-of-date, or missing data.

The SQL part of MySQL stands for Structured Query Language. SQL is the most common standardized language used to access databases. Depending on your programming environment, you might enter SQL directly (for example, to generate reports), embed SQL statements into code written in another language, or use a language-specific API that hides the SQL syntax.

SQL is defined by the ANSI/ISO SQL Standard. The SQL standard has been evolving since 1986 and several versions exist. In this manual, SQL-92 refers to the standard released in 1992, SQL:1999 refers to the standard released in 1999, and SQL:2003 refers to the current version of the standard.

We use the phrase the SQL standard to mean the current version of the SQL Standard at any time. MySQL software is Open Source. Open-Source means that it is possible for anyone to use and modify the software. Anybody can download the MySQL software from the Internet and use it without paying anything. If you wish, you may study the source code and change it to suit your needs.

The MySQL database software use the GPL (GNU General Public License) , <http://www.fsf.org/licenses/>, to define what you may and may not do with the software in different situations. If you feel uncomfortable with the GPL need to embed MySQL code into a commercial application, you can buy a commercially licensed version from us.

The MySQL Database Server is very fast, reliable, scalable, and easy to use. If that is what you are looking for, you should give it a try. MySQL Server can run comfortably on a desktop or laptop, alongside your other applications, web servers, and so on requiring little or no attention. If you dedicate an entire machine to MySQL, you can adjust the settings to take advantage of all the memory, CPU power, and I/O capacity available. MySQL can also scale up to clusters of machines, networked together.

MySQL Server was originally developed to handle large databases much faster than existing solutions and has been successfully used in highly demanding production environments for several years. Although under constant development, MySQL Server today offers a rich and useful set of functions. Its connectivity, speed, and security make MySQL Server highly suited for accessing databases on the Internet. MySQL Server works in client/server or embedded systems.

The MySQL Database Software is a client/server system that consists of a multi-threaded SQL server that supports different backends, several different client programs and libraries, administrative tools, and a wide range of application programming interfaces (APIs). We also provide MySQL Server as an embedded multi-threaded library that you can link into your application to get a smaller, faster, easier-to-manage standalone product.

A large amount of contributed MySQL software is available. MySQL Server has a practical set of features developed in close cooperation with our users. It is very likely that your favourite application or language supports the MySQL Database Server.

2.6.1.3. WAMP SERVER

WAMP Server is a Windows web development environment. It allows you to create web applications with Apache2, PHP and a MySQL database. Alongside, PhpMyAdmin allows you to manage easily your databases. WAMP Server refers to a software stack for the Microsoft Windows operating system, created by Romain Bourdon and consisting of the Apache web server, Open SSL for SSL support, MySQL database and PHP programming language. WAMP Server is a Web development platform on Windows that allows you to create dynamic Web applications with Apache2, PHP, MySQL and MariaDB. WampServer automatically installs everything you need to intuitively develop Web applications. You will be able to tune your server without even

touching its setting files. Best of all, WampServer is available for free (under GPML license) in both 32 and 64-bit versions. Wamp server is not compatible with Windows XP, SP3, or Windows Server 2003. WAMP Server's functionalities are very complete and easy to use so we won't explain here how to use them.

With a left click on WAMP Server's icon, you will be able to:

- Manage your Apache and MySQL services
- Switch online/offline (give access to everyone or only localhost)
- Install and switch Apache, MySQL and PHP releases
- Manage your server's settings
- Access your logs
- Access your settings files
- Create alias

2.6.1.4 VISUAL STUDIO

Visual Studio Code is a source-code editor that can be used with a variety of programming languages, including C, C#, C++, FORTRAN, Go, Java, JavaScript, Node.js, Python, Rust, and Julia. It is based on the Electron framework, which is used to develop Node.js web applications that run on the Blink layout engine. Visual Studio Code employs the same editor component (code named "Monaco") used in Azure DevOps.

Out of the box, Visual Studio Code includes basic support for most common programming languages. This basic support includes syntax highlighting, bracket matching, code folding, and configurable snippets. Visual Studio Code also ships with Intelligent for JavaScript, Type Script, JSON, CSS, and HTML. Support for additional languages can be provided by freely available extensions on the VS Code Marketplace. Instead of a project system, it allows users to open one or more directories, which can then be saved in work spaces for future reuse. This allows it to operate as a language agnostic code editor for any language. It supports many programming languages and a set of features that differs per language. Unwanted files and folders can be excluded

from the project tree via the settings. Many Visual Studio Code features are not exposed through menus or the user interface but can be accessed via the command palette.

2.6.1.5 WINDOWS 11

Windows 11 is the latest version of the Windows operating system, released by Microsoft in October 2021. It's a significant update that offers a refreshed design, new features, and improved performance. It introduces a modern and streamlined design, with a focus on simplicity and ease of use. It is designed to be faster and more efficient, with improved startup times, browsing, and overall system performance. It includes advanced security features, such as improved biometric login, enhanced data protection, and better malware prevention. A new widget panel provides quick access to news, weather, and other information. Teams is now integrated directly into the taskbar, making it easier to stay connected with colleagues and friends. It allows users to run Android apps directly on the desktop. It includes features like Auto HDR, Variable Refresh Rate, and Direct Storage for enhanced gaming performance. It introduces new accessibility features, such as improved eye care, better captioning, and enhanced focus assist. It requires a 64-bit CPU, 4GB of RAM, and a compatible GPU. Eligible Windows 10 devices can upgrade to Windows 11 for free. Overall, Windows 11 offers a fresh and improved user experience, with a focus on performance, security, and ease of use.

Windows 11 features a refreshed design with a new logo, icons, and colour scheme. The interface is more streamlined, with a focus on simplicity and ease of use. It is designed to be faster and more efficient, with improved startup times, browsing, and overall system performance. It includes advanced security features, such as improved biometric login, enhanced data protection, and better malware prevention. It allows users to run Android apps directly on the desktop. It comes in various editions, including Home, Pro, Enterprise, and Education. Navigate Windows without a mouse. Screen reader for visually impaired users. Zoom in and out of desktop and apps. Support for closed captions in videos. Support for high-resolution gaming.

2.6.1.6 MICROSOFT WORD

Microsoft Word (or simply Word) is a word processor developed by Microsoft. It was first released on October 25, 1983 under the name Multi-Tool Word for Xenix systems. Subsequent versions were later written for several other platforms including IBM PCs running DOS (1983), Apple Macintosh running the Classic Mac OS (1985), AT&T Unix PC (1985), Atari ST (1988), OS/2 (1989), Microsoft Windows (1989), SCO Unix (1994), and macOS (formerly OS X; 2001).

Commercial versions of Word are licensed as a standalone product or as a component of Microsoft Office, Windows RT or the discontinued Microsoft Works suite. Unlike most MS-DOS programs at the time, Microsoft Word was designed to be used with a mouse. Advertisements depicted the Microsoft Mouse, and described Word as a WYSIWYG, windowed word processor with the ability to undo and display bold, italic, and underlined text, although it could not render fonts. It was not initially popular, since its user interface was different from the leading word processor at the time, WordStar. However, Microsoft steadily improved the product, releasing versions 2.0 through 5.0 over the next six years. In 1985, Microsoft ported Word to the classic Mac OS (known as Macintosh System Software at the time). This was made easier by Word for DOS having been designed for use with high-resolution displays and laser printers, even though none were yet available to the general public. Following the precedents of Lisa Write and MacWrite, Word for Mac OS added true WYSIWYG features. It fulfilled a need for a word processor that was more capable than MacWrite. After its release, Word for Mac OS's sales were higher than its MS-DOS counterpart for at least four years.

2.6.2 HARDWARE REQUIREMENTS

The selection of hardware configuring is a very task related to the software development, particularly inefficient RAM may affect adversely on the speed and corresponding on the efficiency of the entire system. The processor should be powerful to handle all the operations. The hard disk should have the sufficient to solve the database and the application.

Hardware used for development:

CPU	: Intel i5 Processor
Memory	: 4 GB
Cache	: 6 MB
Hard Disk	: 1 TB
Monitor	: 15.6" Monitor
Keyboard	: Standard 108 keys Enhanced Keyboard
Mouse	: Optical Mouse

Minimum Hardware Required for Implementation:

CPU	: Pentium IV Processor
Memory	: 256MB Above
Cache	: 512 KB Above
Hard Disk	: 20 GB Above
Monitor	: Any
Keyboard	: Any
Mouse	: Any

3.SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

A system architecture or system's architecture is the conceptual model that defines the structure, behaviour, and more views of a system. An architecture description is a formal description and representation of a system, organized in a way that supports reasoning about the structures of the system,

System architecture can comprise system components, the externally visible properties of those components, the relationships (e.g. the behaviour) between them. It can provide a plan from which products can be procured, and systems developed, that will work together to implement the overall system. There have been efforts to formalize languages to describe system architecture; collectively these are called architecture description languages (ADLs).

The system architecture can best be thought of as a set of representations of an existing (or to be created) system. It is used to convey the informational content of the elements comprising a system, the relationships among those elements, and the rules governing those relationships. The architectural components and set of relationships between these components that architecture describes may consist of hardware, software, documentation, facilities, manual procedures, or roles played by organizations or people. System architecture is primarily concerned with the internal interfaces among the system's components or subsystems, and the interface between the system and its external environment, especially the user.

The structural design reduces complexity, facilitates change and result in easier implementation by encouraging parallel development of different parts of the system.

The procedural design transforms structural elements of program architecture into a procedural description of software components. The architectural design considers architecture as the most important functional requirement. The system is based on the three-tier architecture.

The first level is the user interface (presentation logic), which displays controls, receives and validates user input. The second level is the business layer (business logic) where the application specific logic takes place. The third level is the data layer where the application information is stored in files or database. It contains logic about to retrieve and update data. The important feature about the three-tier design is that information only travels from one level to an adjacent level.

3.2 MODULE DESIGN

Modular programming is a software design technique that emphasizes separating the functionality of a program into independent, interchangeable modules, such that each contains everything necessary to execute only one aspect of the desired functionality.

Conceptually, modules represent a separation of concerns, and improve maintainability by enforcing logical boundaries between components.

Modules of this project:

1. Registration

This module manages all registration processes within the system. It includes registration performed by students and companies through the platform. Students register by providing personal and academic details required for applying to internships, while companies register through the guest page by submitting organization details for verification.

The company registration request is reviewed and approved or rejected by the admin to ensure authenticity and reliability of internship providers. Only approved companies are allowed to access system functionalities such as posting internships and managing applications. The admin also manages user roles and controls access permissions within the system.

All registration details are securely stored in the database, ensuring data integrity, privacy, and controlled access to the InternHub platform.

2. Authentication

This module manages the authentication process of all users within the InternHub system. It ensures secure access to the platform by validating login credentials of students, companies, and admin using authorized usernames and passwords. Authentication helps in preventing unauthorized access and protects sensitive system data.

After successful registration, students and approved companies can log in to the system using their credentials. Companies are permitted to access the system only after admin approval of their registration request. The admin can securely log in to monitor and manage overall system activities.

The module also provides facilities for session management, password verification, and secure logout options. All authentication details are maintained securely to ensure privacy, system security, and controlled access to InternHub functionalities.

3. Activities

This module handles the core functional operations of the InternHub system. Students can search companies, view available internships, access internship details such as duration, syllabus, and fees, and apply for internships through the platform. Companies can add internship programs, manage internship details, and review applications submitted by students.

Companies are responsible for approving or rejecting student applications based on eligibility requirements. The admin monitors company activities, internship postings, applications, and overall system operations to ensure proper functioning of the platform.

This module ensures smooth interaction between students, companies, and admin while supporting an efficient and transparent internship management process within the InternHub system.

4. Payment

This module manages all payment-related activities within the InternHub system. After the company approves a student's internship application, the student is allowed to proceed with the payment process through the platform. The module ensures secure and reliable payment transactions for internship enrollment.

Students can make payments for selected internships, and the system records payment details for future reference. A predefined percentage of the payment (10%) is allocated to the admin as a service commission, while the remaining amount is transferred to the respective company providing the internship.

The admin and companies can monitor payment status and transaction details through the system. This module ensures transparency, accuracy, and secure handling of financial transactions within the InternHub platform.

5. Report

This module manages the generation and maintenance of reports within the InternHub system. The admin can view and generate reports related to registered students, approved companies, internship postings, applications, and payment transactions. These reports help in monitoring overall system performance and activities.

Companies can view reports related to internship applications, approved students, and payment details associated with their internship programs. Students can also view their application history and internship status through the system.

The reports module provides organized and accurate information that supports decision-making, performance analysis, and effective management of internship activities within the InternHub platform.

3.3 DATABASE DESIGN

A database is a collection of interrelated data stored with minimum redundancy to serve many users quickly and efficiently. The general objective is to make information access easy, quick, inexpensive and flexible for the users. The general theme behind a database is to integrate all information. Database design is recognized as a standard of management information system and is available virtually for every computer system.

In database design several specific objectives are considered:

- Ease of learning and use
- Controlled redundancy
- Data independence
- More information at low cost
- Accuracy and integrity
- Recovery from failure
- Privacy and security
- Performance

A database is an integrated collection of data and provides centralized access to the data. Usually, the centralized data managing the software is called RDBMS. The main significant difference between RDBMS and other DBMS is the separation of data as seen by the program and data has in direct access to stores device. This is the difference between logical and physical data.

3.3.1 Normalization

Designing a database is complete task and the normalization theory is a useful aid in the design process. The process of normalization is concerned with transformation of conceptual schema into computer representation form. There will be need for most databases to grow by adding new attributes and new relations. The data will be used in new ways. Tuples will be added and deleted. Information stored may undergo updating

also. New association may also be added. In such situations the performance of a database is entirely depend upon its design. A bad database design may lead to certain undesirable things like:

- Repetition of information
- Inability to represent certain information
- Loss of information

To minimize these anomalies, Normalization may be used. If the database is in a normalized form, the data can be growing without, in most cases, forcing the rewriting application programs. This is important because of the excessive and growing cost of maintaining an organization's application programs and its data from the disrupting effects of database growth. As the quality of application programs increases, the cost of maintaining the without normalization will rise to prohibitive levels. A normalized database can also encompass many related activities of an organization thereby minimizing the need for rewriting the applications of programs. Thus, normalization helps one attain a good database design and there by ensures continued efficiency of database.

Normalization theory is built around the concept of normal forms. A relation is said to be in normal form if it satisfies a certain specified set of constraints. For example, a relation is said to be in first normal form (1NF) if it satisfies the constraint that it contains atomic values only. Thus, every normalized relation is in 1NF. Numerous normal forms have been defined. Codd defined the first three normal forms.

All normalized relations are in 1NF, some 1NF relations are also in 2NF and some 2NF relations are also in 3NF. 2NF relations are more desirable than 1NF and 3NF are more desirable than 2NF. That is, the database designer should prefer 3NF than 1NF or 2NF. Normalization procedure states that a relation that is in some given normal form can be converted into a set of relations in a more desirable form. We can define this procedure as the successive reduction of a given collection of relations to some more desirable form. This procedure is reversible. That is, it is always possible to take the output from the procedure and convert them back into input. In this process, no information is lost. So it is also called "no loss decomposition".

First Normal Form

A relation is in first normal form (1NF) if and all its attributes are based on single domain. The objective of normalizing a table is to remove its repeating groups and ensure that all entries of the resulting table have at most single value. **Second Normal**

Form

A table is said to be second Normal Form (2NF), when it is in 1NF and every attribute in record is functionally dependent upon the whole key, and not just a part of the key.

Third Normal Form

A table is in third Normal Form (3NF), when it is in 2NF and every non-key attribute is functionally dependent on just the primary key.

3.3.2 Table Structure

Table is a collection of complete details about a particular subject. These data are saved in rows and Columns. The data of each Row are different units. Hence, rows are called RECORDS and Columns of each row are called FIELDS.

Data is stored in tables, which is available in the backend the items and data, which are entered in the input, form id directly stored in this table using linking of database. We can link more than one table to input forms. We can collect the details from the different tables to display on the output. There are mainly 10 tables in the project. They are,

1. tbl_login
2. tbl_location
3. tbl_district
4. tbl_category
5. tbl_companytype
6. tbl_company
7. tbl_payment
8. tbl_internship
9. tbl_request_internship
10. tbl_student

1. Table: tbl_login

Description: This table is used to store login details of the stakeholders.

Table 3.1 tbl_login

Field Name	Data Type	Constraint	Description of field
loginid	Int	Primary key	This field uniquely identifies the admin, with values being automatically generated.
username	Varchar(20)	Not null	This field stores the username.
password	Varchar(20)	Not null	This field stores the password.
role	Varchar(20)	Not null	This field stores the role of the system user.
status	Varchar(20)	Not null	This field stores the status of the system user.

2. Table: tbl_district

Description: This table is used to store district information.

Table 3.2 tbl_district

Field name	Data type	Constraints	Description
district_id	int	Primary key	This field is used to uniquely identify district, and the values are automatically generated.
district_name	Varchar(30)	Not null	This field stores the name of the district.

3. Table: **tbl_location**

Description: This table is used to store location information.

Table 3.3 tbl_location

Field name	Data type	Constraints	Description
location_id	int	Primary key	This field is used to uniquely identify location, and the values are automatically generated.
location_name	Varchar(30)	Not null	This field stores the name of the location.
district_id	Int	Foreign Key	This field is used to refer the table tbl_district.

4. Table: **tbl_category**

Description: This table is used to store category information.

Table 3.4 tbl_category

Field name	Data type	Constraints	Description of field
categoryid	Int	Primary key	This field uniquely identifies the categories, with values being automatically generated.
categoryname	Varchar (40)	Not null	This field stores the categories name.

5. Table: **tbl_company**

Description: This table is used to store company information.

Table 3.5 tbl_company

Field name	Data type	Constraints	Description of field
companyid	Int	Primary key	This field uniquely identifies the officer, with values being automatically generated.
companyname	Varchar (150)	Not null	This field stores the officer name.
logo	Varchar (150)	Null	This field stores the company logo.
email	Varchar (30)	Not null	This field is used to store the email of the company.
contact	Bigint	Not null	This field is used to store the contact of the company.
loginid	Int	Foreign Key	This field is used to refer the table tbl_login.
locationid	Int	Foreign Key	This field is used to refer the table tbl_location.
companytypeid	Int	Foreign Key	This field is used to refer the table tbl_companytype.
website	Varchar (200)	Not null	This field stores the website of the company.
regdate	Date	Null	This field is used to store the company's registration date.

6. Table: tbl_companytype

Description: This table is used to store company type information.

Table 3.6 tbl_companytype

Field name	Data type	Constraints	Description of field
companytypeid	Int	Primary key	This field uniquely identifies the companytype, with values being automatically generated.
companytype name	Varchar (40)	Not null	This field stores the companytype name.

7. Table: tbl_payment

Description: This table is used to store payment information

Table 3.7 tbl_payment

Field name	Data type	Constraints	Description of field
paymentid	Int	Primary key	This field uniquely identifies the payment, with values being automatically generated.
requestid	Int	Foreign Key	This field is used to refer the table tbl_request_internship
totalamount	Varchar (200)	Not null	This field stores the amount.
paymentdate	Date	Not null	This field is used to store the payment date.
camount	decimal (200)	Not null	This field stores the camount.
status	Varchar (50)	Not null	This field stores the payment status.

8. Table: tbl_internship

Description: This field is used to store the internship information

Table 3.8 tbl_internship

Field name	Data type	Constraints	Description of field
internshipid	Int	Primary key	This field uniquely identifies the internship, with values being automatically generated.
name	Varchar (50)	Not null	This field stores the internship name.
syllabus	Varchar (30)	Not null	This field is used to store the syllabus of the internship.
duration	Bigint	Not null	This field is used to store the duration of internship.
amount	decimal (200)	Not null	This field stores the amount.
description	Varchar (200)	Not null	This field stores the internship description.
photo	Varchar(150)	Not null	This field is used to store the photo.
companyid	Int	Foreign Key	This field is used to refer the table tbl_company.

9 Table: tbl_request_internship

Description: This table is used to store internship request

Table 3.9 tbl_request_internship

Field name	Data type	Constraints	Description of field
requestid	Int	Primary key	This field uniquely identifies the internship request, with values being automatically generated.
requestdate	Date	Not null	This field is used to store the internship request date.
startdate	Date	Not null	This field is used to store the internship start date.
internshipid	Int	Foreign Key	This field is used to refer the table tbl_internship.
studentid	Int	Foreign Key	This field is used to refer the table tbl_student.
status	Varchar (50)	Not null	This field stores the request status.

10. Table: tbl_student

Description: This table is used to store student information.

Table 3.10 tbl_student

Field name	Data type	Constraints	Description
studentid	Int	Primary key	This field uniquely identifies the student id, with values being automatically generated.
name	Varchar (50)	Not null	This field stores the student name.
photo	Varchar(150)	Null	This field is used to store the photo.
email	Varchar (30)	Not null	This field is used to store the email of the student.
contact	Bigint	Not null	This field is used to store the contact of the student.
dateofbirth	Date	Null	This field is used to store the date of birth of the student.
collegename	Varchar (50)	Not null	This field stores the college name.
course	Varchar (50)	Not null	This field stores the course name.
semester	Varchar (50)	Not null	This field stores the semester.
locationid	Int	Foreign Key	This field is used to refer the table tbl_location.
loginid	Int	Foreign Key	This field is used to refer the table tbl_login.

3.3.3 Data Flow Diagram

3.3.3.1. Introduction to Data Flow Diagrams

Data Flow Diagram is a network that describes the flow of data and processes that change, or transform, data throughout the system. This network is constructed by use a set of symbols that do not imply a physical implementation. It is a graphical tool for structured analysis of the system requirements. DFD models a system by using external entities from which data flows to a process, which transforms the data and creates, output-data-flows which go to other processes or external entities or files. Data in files may also flow to processes as inputs.

There are various symbols used in a DFD. Bubbles represent the processes. Named arrows indicate the data flow. External entities are represented by rectangles. Entities supplying data are known as sources and those that consume data are called sinks. Data are stored in a data store by a process in the system. Each component in a DFD is labelled with a descriptive name. Process names are further identified with a number.

The Data Flow Diagram shows the logical flow of a system and defines the boundaries of the system. For a candidate system, it describes the input (source), outputs (destination), database (files) and procedures (data flow), all in a format that meet the user's requirements.

The main merit of DFD is that it can provide an overview of system requirements, what data a system would process, what transformations of data are done, what files are used, and where the results flow.

This network is constructed by use a set of symbols that do not imply a physical implementation. It is a graphical tool for structured analysis of the system requirements. DFD models a system by using external entities from which data flows to a process, which transforms the data and creates, output-data-flows which go to other processes or external entities or files.

External entities are represented by rectangles. Entities supplying data are known as sources and those that consume data are called sinks. Data are stored in a data store by a process in the system. It is a graphical tool for structured analysis of the system requirements. DFD models a system by using external entities from which data flows to a process, which transforms the data and creates, output-data- flows which go to other processes or external entities or files. Data in files may also flow to processes as inputs.

Rules for constructing a Data Flow Diagram

1. Arrows should not cross each other.
2. Squares, circles and files must bear names.
3. Decomposed data flow squares and circles can have same time.
4. Choose meaningful names for data flow.
5. Draw all data flows around the outside of the diagram.

Basic Data Flow Diagram Symbols

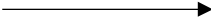
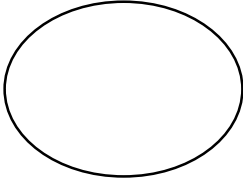
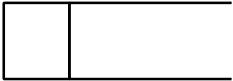
	A data flow is a route, which enables packets of data to travel from one point to another. Data may flow from a source to a process and from data store or process. An arrow line depicts the flow, with arrow head pointing in the direction of the flow.
	Circles stand for process that converts data in to information. A process represents transformation where incoming data flows are changed into outgoing data flows.
	A data store is a repository of data that is to be stored for use by a one or more process may be as simple as buffer or queue or sophisticated as relational database. They should have clear names. If a process merely uses the content of store and does not alter it, the arrowhead goes only from the store to the process. If a process alters the details in the store, then a doubleheaded arrow is used.
	A source or sink is a person or part of an organization, which enters or receives information from the system, but is considered to be outside the contest of data flow model.

Table 3.11 Data Flow Diagram Symbols

3.3.3.2 Data Flow Diagram

Each component in a DFD is labelled with a descriptive name. Process name are further identified with number. Context level DFD is draw first. Then the process is decomposed into several elementary levels and is represented in the order of importance. A DFD describes what data flow (logical) rather than how they are processed, so it does not depend on hardware, software, and data structure or file organization. A DFD methodology is quite effective; especially when the required design.

ZEROTH LEVEL OR CONTEXT LEVEL DIAGRAM

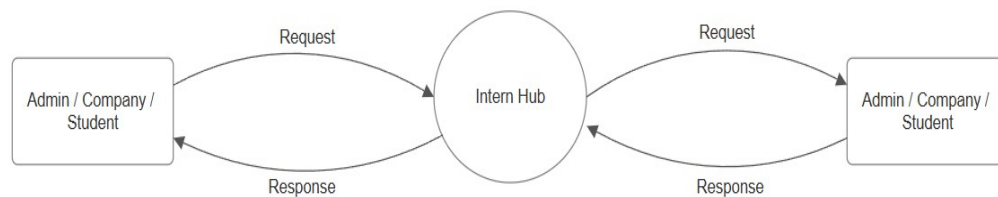


fig 3.1 Zeroth level DFD for Internhub

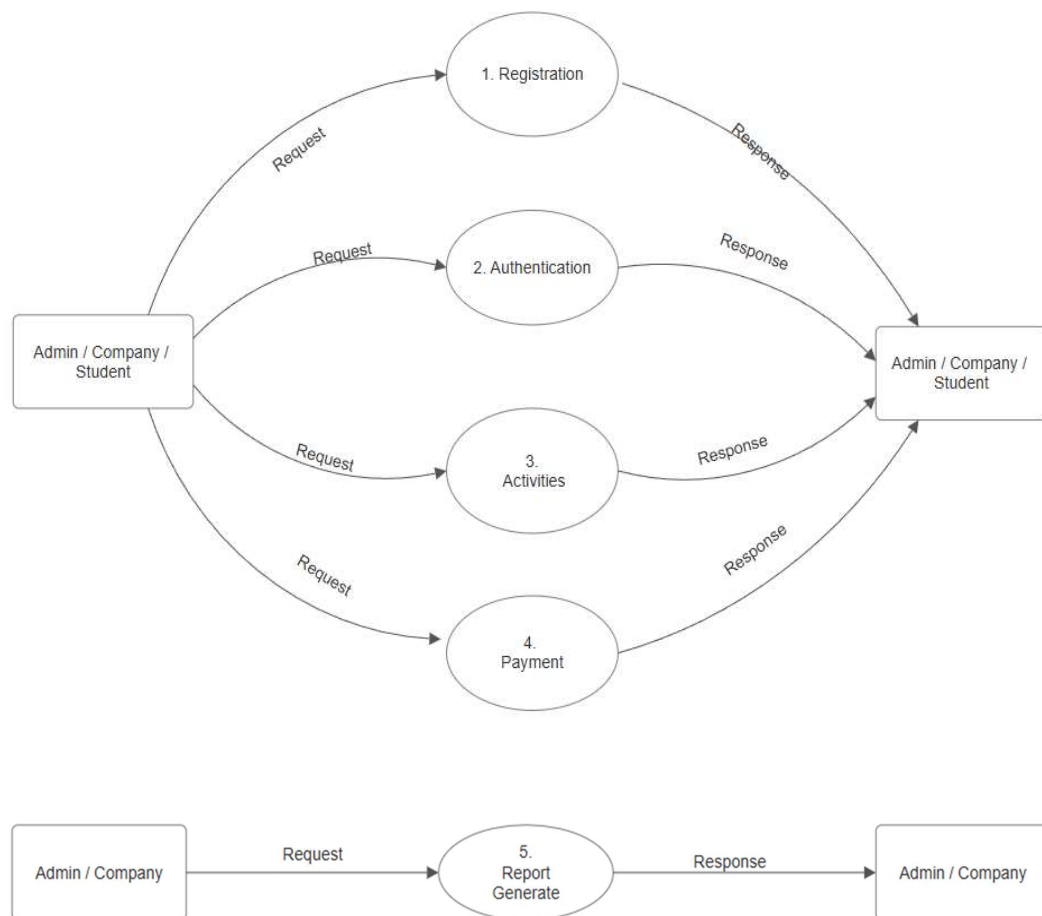
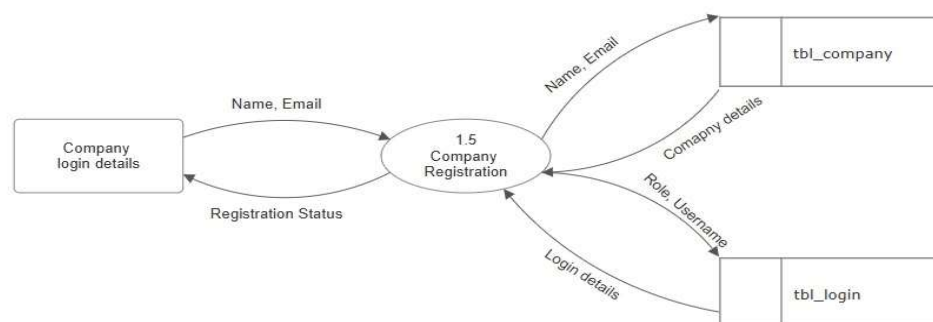
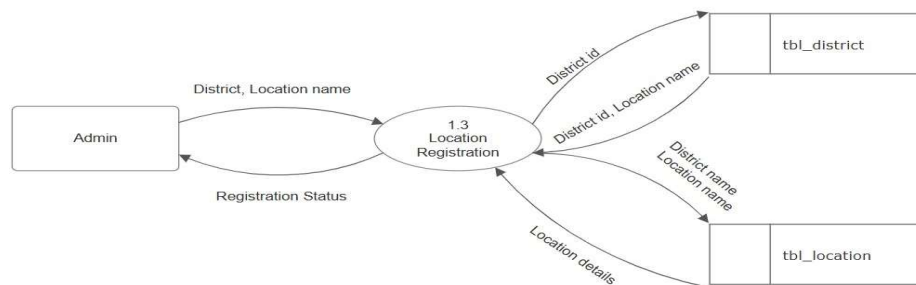
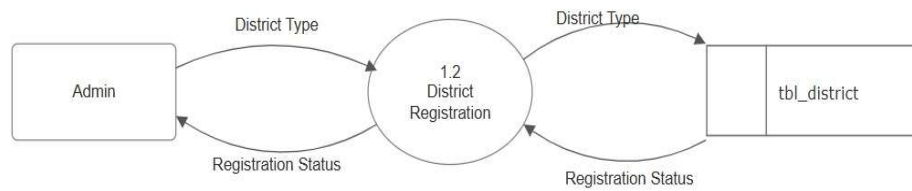
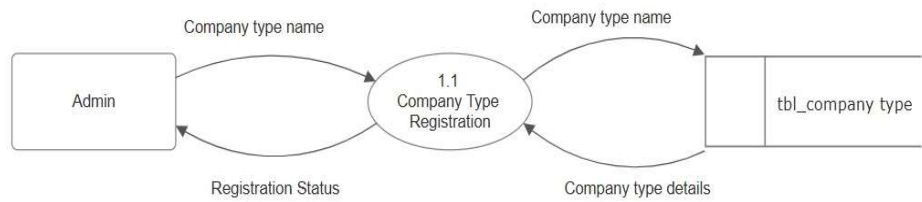
FIRST LEVEL DIAGRAM

fig 3.2 First level DFD for InternHub

SECOND LEVEL DIAGRAM



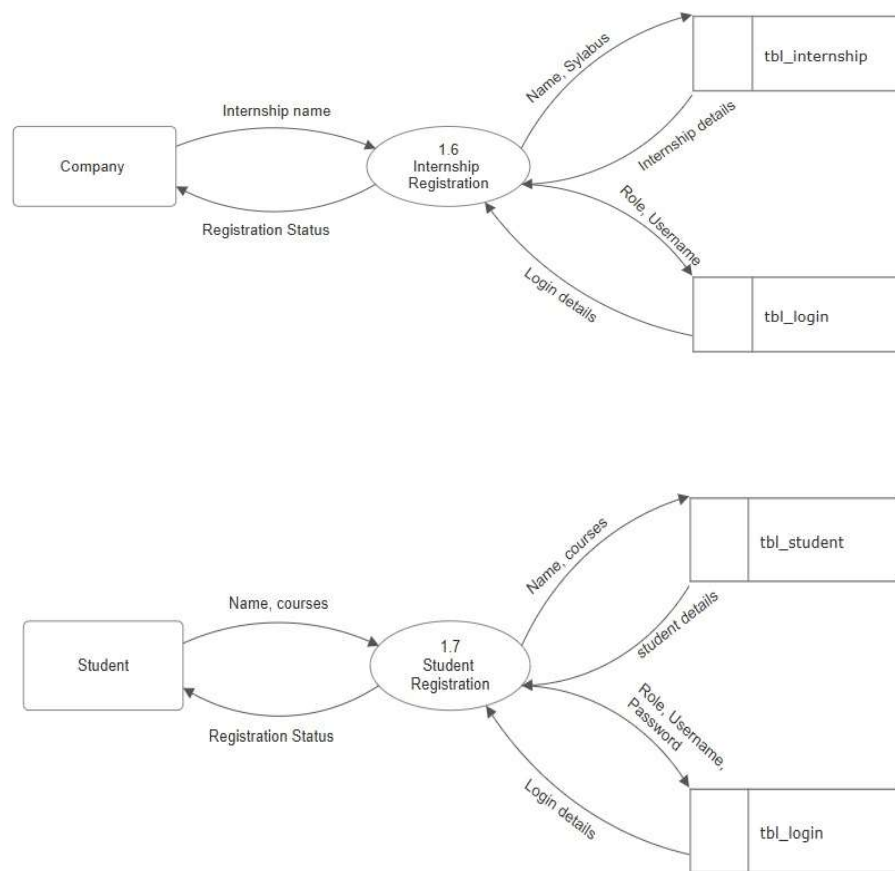


Fig. 3.3 Second level DFD for Registration

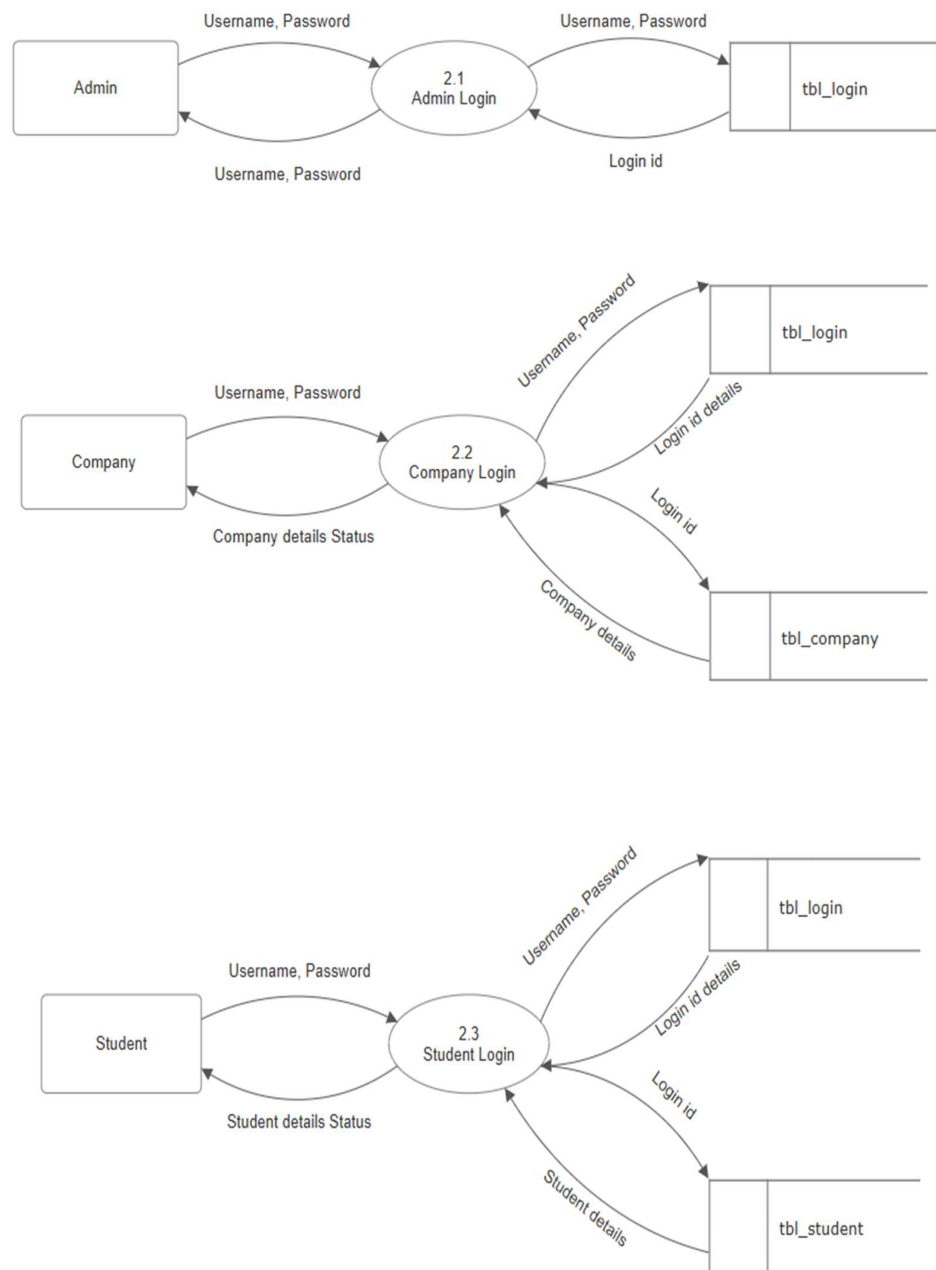
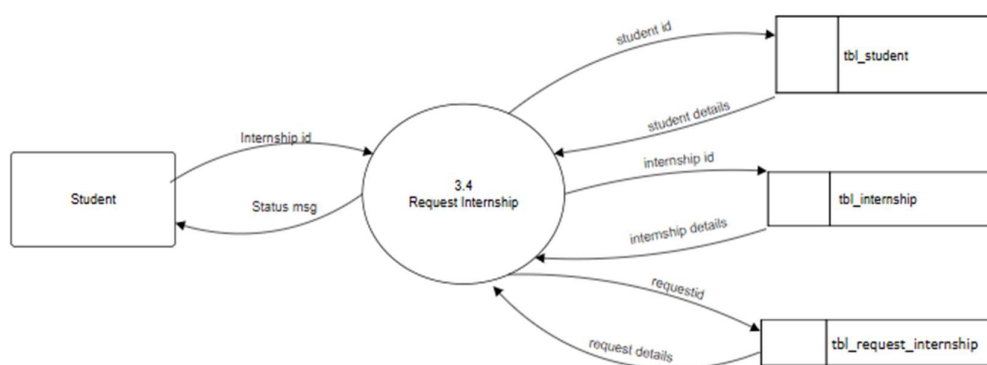
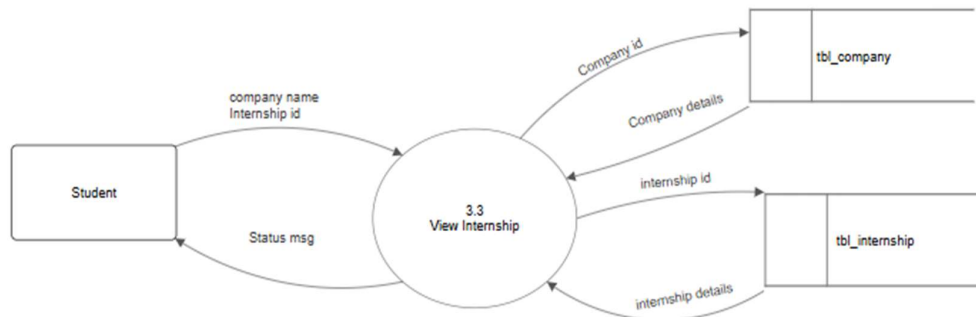
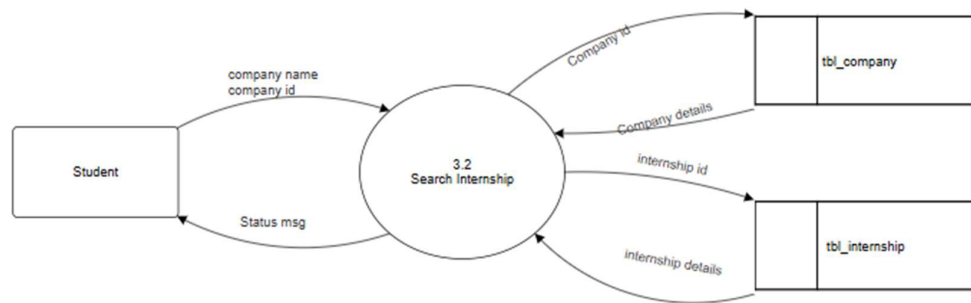
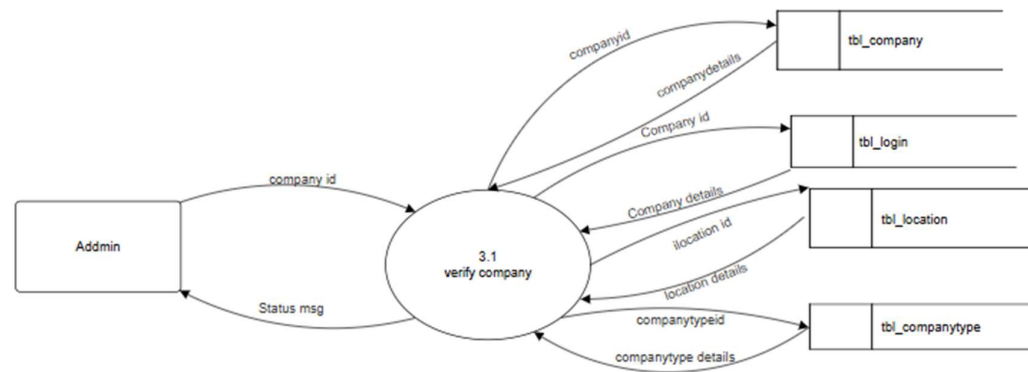


Fig. 3.4 Second Level DFD for Authentication



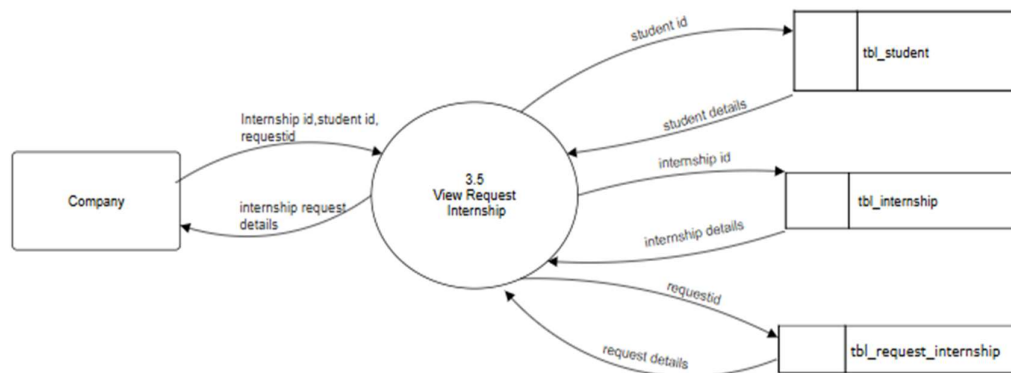


Fig. 3.5 Second Level DFD for Activity

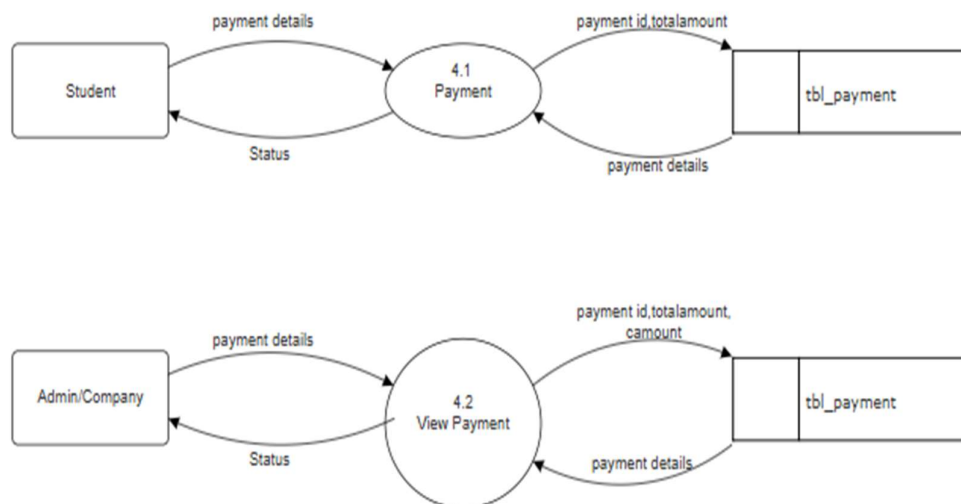


Fig. 3.6 Second Level DFD for Payment

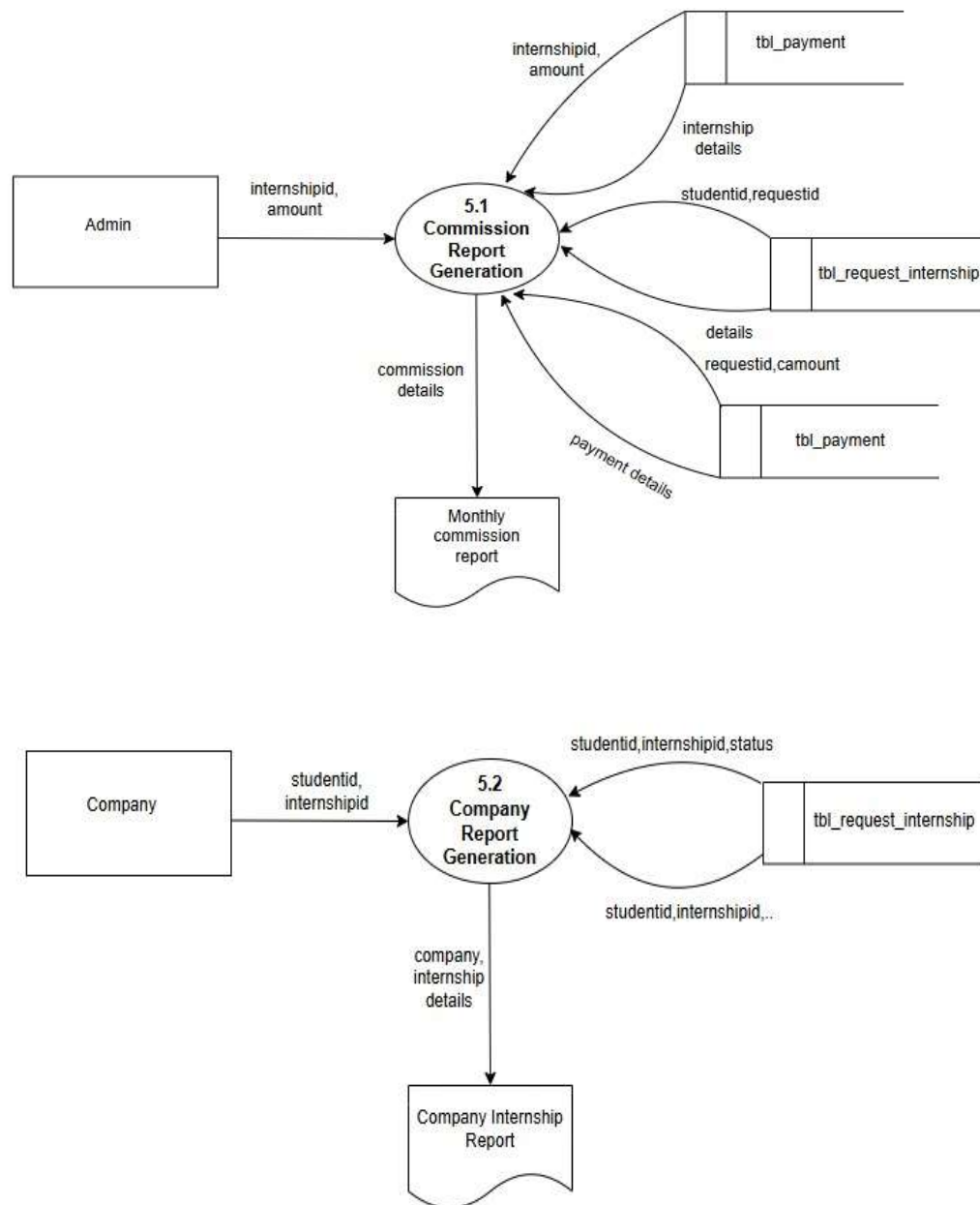


Fig. 3.7 Second Level DFD for Report

3.4 INTERFACE DESIGN

These modules can apply to hardware, software or the interface between a user and a machine. An example of a user interface could include a GUI, a control panel for a nuclear power plant, or even the cockpit of an aircraft. In systems engineering, all the inputs and outputs of a system, subsystem, and its components are listed in an interface control document often as part of the requirements of the engineering project. The development of a user interface is a unique field.

3.4.1 User Interface Screen Design

The user interface design is very important for any application. The interface design describes how the software communicates within itself, to system that interpreted with it and with humans who use it. The input design is the process of converting the user-oriented inputs into the computer-based format. The data is fed into the system using simple inactive forms. The forms have been supplied with messages so that the user can enter data without facing any difficulty. They data is validated wherever it requires the project. This ensures that only the correct data have been incorporated into system. The goal of designing input data is to make the automation as easy and free from errors as possible. For providing a good input design for the application easy data input and selection features are adopted. The input design requirements such as user friendliness, consistent format and interactive dialogue for giving the right messages and help for the user at right are also considered for development for this project.

Input Design is a part of the overall design. The input methods can be broadly classified into batch and online. Internal controls must be established for monitoring the number of inputs and for ensuring that the data are valid. The basic steps involved in input design are:

- Review input requirements.
- Decide how the input data flow will be implemented.
- Decide the source document.
- Prototype on line input screens.
- Design the input screens

This is an input form used for creating a profile for a new student in the InternHub system. It contains textboxes for entering student details such as name, address, contact number, email, educational qualification, course, institution name, username, and password. The form also includes fields required for internship application purposes. After clicking the submit button, the entered details will be stored in the system database and a registration successful message will be displayed. The registered student will then be redirected to the login page.

The username and password created during registration help the student to log in to the system after logout. Each student must fill all mandatory details provided in the form to successfully register and access the InternHub platform. Each field contains appropriate labels indicating the type of information to be entered. Textboxes also include placeholders that guide users in entering valid data.

The form includes a submission button that allows users to send the entered information to the database table. The data entered must satisfy validation conditions according to each field type. All labels and input fields are properly aligned to maintain a clear and user-friendly interface design.

3.4.2 Output Design

A quality output is one, which meets the requirements of end user and presents the information clearly. In any system result of processing are communicated to the user and to the other system through outputs. In the output design it is determined how the information is to be displayed for immediate need. It is the most important and direct source information is to the user. Efficient and intelligent output design improves the system's relationships with the user and helps in decision -making. The objective of the output design is to convey the information of all the past activities, current status and to emphasis important events. The output generally refers to the results and information that is generated from the system. Outputs from computers are required primarily to communicate the results of processing to the users. Output also provides a means of storage by copying the results for later reference in consultation. There is a chance that some of the end users will not actually operate the input data or information through workstations, but will see the output from the system.

Two phases of the output design are:

1. Output Definition

2. Output Specification

Output Definition takes into account the type of output contents, its frequency and its volume, the appropriate output media is determined for output. Once the media is chosen, the detail specification of output documents are carried out. The nature of output required from the proposed system is determined during logical design stage. It takes the outline of the output from the logical design and produces output as specified during the logical design phase.

In a project, when designing the output, the system analyst must accomplish the following:

- Determine the information to present.
- Decide whether to display, print, speak the information and select the output medium.
- Arrange the information in acceptable format.
- Decide how to distribute the output to the intended receipt.
- Thus, by following the above specifications, a high-quality output can be generated.

4. IMPLEMENTATION

Implementation is the stage of the project when the theoretical design is turned into a working system. The implementation stage is a systems project in its own right. It includes careful planning, investigation of current system and its constraints on implementation, design of methods to achieve the changeover, training of the staff in the changeover procedure and evaluation of changeover method.

4.1 CODING STANDARDS

Python Django, like PHP, adheres to specific coding standards to ensure consistency and clarity in codebases. With developers worldwide contributing to projects, diverse coding styles can lead to confusion and hinder collaboration. Establishing and adhering to coding standards is crucial for code readability, maintainability, and future reference. Consistent formatting acts as a blueprint, making code easier to understand, navigate, and modify. These standards not only enhance code quality but also align with industry best practices. Below are essential guidelines to uphold Python Django coding standards

- 1. Python Tags:** Python doesn't use tags to delimit code like PHP does. Instead, it uses indentation to denote code blocks
- 2. Commenting:** Python primarily uses `#` for both single-line and multi-line comments.
- 3. Line Length and Indentation:** PEP 8 recommends a maximum line length of 79 characters. Indentation in Python is done with 4 spaces.
- 4. Structuring Control Flow Statements:** Python control flow statements like `if`, `for`, `while`, etc., have a similar structure to PHP. The spacing convention remains consistent.

Example:

- 5. Function Calls:** While writing a function call statement, there must be no space between the function name and the opening parenthesis.

Example: `filter_none edit play_arrow`

`brightness_4 n = 5 if n > 0:`

```
print("Positive")
```

```
elif n < 0:
```

```
print("Negative")
```

```
else:
```

```
print("Zero")
```

Output:

Positive

6. Naming Variables: Python follows the convention of using lowercase letters to name variables, with words separated by underscores. There's no specific convention for static or global variable names like in PHP.

7. Block alignment: Python also emphasizes code readability and encourages aligning blocks and curly braces.

8. Short Functions: Python encourages writing functions and methods that are concise and focused on a single task, similar to PHP.


```
padding: 50px 20px;
}

/* Card Design */
.card{
    border: none;
    border-radius: 18px;
    box-shadow: 0 20px 45px rgba(0,0,0,0.08);
}

.card-body{
    padding: 45px;
}

/* Title */
.card-title{
    font-size: 28px;
    font-weight: 600;
    color: #1e3a8a;
    text-align: center;
    margin-bottom: 30px;
}

.title-line{
    width: 70px;
    height: 4px;
    background: #2563eb;
    margin: 12px auto 30px;
    border-radius: 5px;
}

/* Form Labels */
label{
```

```
font-weight: 500;
color: #334155;
margin-bottom: 6px;
display: block;
}

/* Inputs & Select */
.form-group{
  margin-bottom: 20px;
}

.form-control{
  border-radius: 10px;
  border: 1px solid #cbd5e1;
  padding: 12px 14px;
  font-size: 14px;
  transition: 0.3s ease;
}

.form-control:focus{
  border-color: #2563eb;
  box-shadow: 0 0 0 3px rgba(37,99,235,0.15);
}

/* Buttons */
.btn-primary{
  background: #2563eb;
  border: none;
  padding: 10px 28px;
  border-radius: 8px;
  font-weight: 500;
  transition: 0.3s ease;
}
```

```
.btn-primary:hover{  
    background: #1d4ed8;  
}
```

```
.btn-light{  
    border-radius: 8px;  
    padding: 10px 28px;  
}
```

```
/* Responsive */  
@media(max-width:768px){  
    .card-body{  
        padding: 25px;  
    }  
}  
</style>
```

```
<script>  
$(document).ready(function() {  
    $.ajaxSetup({  
        headers : {"X-CSRFToken" : '{{csrf_token}}'}  
    });
```

```
$('#district').change(function() {  
    var districtid = $(this).val();  
    $.ajax({  
        type: "POST",  
        url : '{% url "filllocation" %}',  
        data : {did: districtid},  
        datatype: "json",  
        success : function(data) {  
            $('#location').empty();
```

```
$('#location').append('<option value="">Select Location</option>');
$.each(data,function(key,val) {
var row = "<option value='"+val.locationid+"'>"+val.locationname+"</option>";
    $('#location').append(row);
});
}
});
});
</script>
```

```
<div class="main-panel">
  <div class="content-wrapper">
    <div class="row justify-content-center">
      <div class="col-md-8 grid-margin stretch-card">
        <div class="card">
          <div class="card-body">

            <h4 class="card-title">Company Registration</h4>
            <div class="title-line"></div>

            <form method="POST" action="{% url 'company_insert' %}"
                    enctype="multipart/form-data">
              {% csrf_token %}

              <!-- Company Name -->
              <div class="form-group">
                <label>Company Name</label>
                <input type="text" class="form-control" name="companyname"
                        placeholder="Enter company name"
                        pattern="[A-Z][a-zA-Z\s]*"
                        title="Name must start with a capital letter"
                        required>
```

```
</div>

<!-- Logo -->
<div class="form-group">
  <label>Logo (JPEG only)</label>
  <input type="file" class="form-control" name="logo"
    accept="image/jpeg" required>
</div>

<!-- Email -->
<div class="form-group">
  <label>Email</label>
  <input type="email" class="form-control" name="email"
    placeholder="Enter email"
    required>
</div>

<!-- Contact -->
<div class="form-group">
  <label>Contact Number</label>
  <input type="text" class="form-control" name="contact"
    placeholder="Enter contact number"
    pattern="[6-9][0-9]{9}"
    title="Enter valid 10-digit mobile number"
    required>
</div>

<!-- Website -->
<div class="form-group">
  <label>Website</label>
  <input type="text" class="form-control" name="website"
    placeholder="Enter website URL">
</div>
```

```
<!-- Company Type -->
<div class="form-group">
  <label>Company Type</label>
  <select class="form-control" name="companytypeid" required>
    <option value="">Select Company Type</option>
    {% for company in companytype %}
      <option value="{{company.companytypeid}}">
        {{company.companytypename}}
      </option>
    {% endfor %}
  </select>
</div>

<!-- District -->
<div class="form-group">
  <label>District</label>
  <select class="form-control" name="districtid" id="district" required>
    <option value="">Select District</option>
    {% for d in district %}
      <option value="{{ d.districtid }}">
        {{ d.districtname }}
      </option>
    {% endfor %}
  </select>
</div>

<!-- Location -->
<div class="form-group">
  <label>Location</label>
  <select class="form-control" name="locationid" id="location" required>
    <option value="">Select Location</option>
    {% for loc in location %}
```

```
<option value="{{ loc.locationid }}">
    {{ loc.locationname }}
</option>
{% endfor %}
</select>
</div>

<!-- Username -->
<div class="form-group">
    <label>Username</label>
    <input type="text" class="form-control" name="username"
        placeholder="Enter username" required>
</div>

<!-- Password -->
<div class="form-group">
    <label>Password</label>
    <input type="password" class="form-control" name="password"
        placeholder="Enter password"
        pattern="^(?=.*[A-Za-z])(?=.*\d)(?=.*[@$!%*?&]).{6,}$"
        title="At least 6 characters with letter, number & special character"
        required>
</div>

<!-- Buttons -->
<div class="text-center mt-4">
    <button type="submit" class="btn btn-primary mr-2">Submit</button>
    <button type="reset" class="btn btn-light">Cancel</button>
</div>

</form>

</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
{% include 'guest/footer.html' %}
```

5. TESTING

Coding conventions are a set of guidelines for a specific programming language that recommend programming style, practices and methods for each aspect of a piece program written in this language. These conventions usually cover file organization, indentation, comments, declarations, statements, white space, naming conventions, programming practices, programming principles, programming rules of thumb, architectural best practices, etc. These are guidelines for software structural quality. Software programmers are highly recommended to follow these guidelines to help improve the readability of their source code and make software maintenance easier.

5.1 TEST CASES

The objective of system testing is to ensure that all individual programs are working as expected, that the programs link together to meet the requirements specified and to ensure that the computer system and the associated clerical and other procedures work together. The initial phase of system testing is the responsibility of the analyst who determines what conditions are to be tested, generates test data, produced a schedule of expected results, runs the tests and compares the computer produced results with the expected results with the expected results. The analyst may also be involved in procedures testing. When the analyst is satisfied that the system is working properly, he hands it over to the users for testing. The importance of system testing by the user must be stressed. Ultimately it is the user must verify the system and give the go-ahead.

During testing, the system is used experimentally to ensure that the software does not fail, i.e., that it will run according to its specifications and in the way users expect it to. Special test data is input for processing (test plan) and the results are examined to locate unexpected results. A limited number of users may also be allowed to use the system so analysts can see whether they try to use it in unexpected ways. It is preferably to find these surprises before the organization implements the system and depends on it. In many organizations, testing is performed by person other than those who write the original programs. Using persons who do not know how certain parts were designed or programmed ensures more complete and unbiased testing and more reliable software. Parallel running is often regarded as the final phase of system testing.

Since the parallel operation of two systems is very demanding in terms of user resources it should be embarked on only if the user is satisfied with the results of testing -- it should not be started if problems are known to exist. Testing is the major quality control measure during software development. Its basic function is to detect errors in the software. Thus, the goal of testing is to uncover requirement design and coding errors in the program.

Testing is the process of correcting a program with the intent of finding an error. Different types of testing are,

1. Unit Testing
2. Integrated Testing
3. Black Box Testing
4. White Box Testing
5. Validation Testing
6. User Acceptance Testing

5.1.1 Unit Testing

In computer programming, unit testing is a method by which individual units of source code, sets of one or more computer program modules together with associated control data, usage procedures, and operating procedures are tested to determine if they are fit for use. In this testing we test each module individually and integrated the overall system. Unit testing focuses verification efforts on the smaller unit of software design in the module. This is also known as module testing. The modules of the system are tested separately. The testing is carried out during programming stage itself. In this testing step each module is found to be working satisfactorily as regards the expected output from the module. There are some validation checks for verifying the data input given by the user which both the form and validity of the entered. It is very easy to find and debug the system.

5.1.4 User Acceptance Testing

Acceptance Testing is a level of the software testing process where a system is tested for acceptability. User Acceptance testing is the software testing process where system tested for acceptability & validates the end-to-end business flow. Such type of testing executed by client in separate environment & confirms whether system meets the requirements as per requirement specification or not.

UAT is performed after System Testing is done and all or most of the major defects have been fixed. This testing is to be conducted in the final stage of Software Development Life Cycle (SDLC) prior to system being delivered to a live environment. UAT users or end users are concentrating on end to end scenarios & typically involves running a suite of tests on the completed system.

User Acceptance testing also known as Customer Acceptance testing (CAT), if the system is being built or developed by an external supplier. The CAT or UAT are the final confirmation from the client before the system is ready for production. The business customers are the primary owners of these UAT tests. These tests are created by business customers and articulated in business domain languages. So ideally it is collaboration between business customers, business analysts, testers and developers. It consists of test suites which involve multiple test cases & each test case contains input data (if required) as well as the expected output. The result of test case is either a pass or fail.

5.2 TEST CASE DOCUMENTS

A test case is a set of conditions or variables under which a tester will determine whether a system under test satisfies requirements or works correctly. The process of developing test cases can also help find problems in the requirements or design an application. A sample of test case document format is given below

Table 5.1 Test Case

Tc no	Test shape	Expected result	Actual result	status	Comment
1	Run application to navigate to add product Screen	Product registration page should be displayed with input field for inserting product details	Product registration page displayed with essential input fields	pass	
2	View page for viewing added products	Registered product view page should be displayed with product details in a tabular form	Registered product view page displayed with product details.	pass	
3	Customer shouldn't place order without filling payment details	While clicking place order button in checkout without filling payment details, a warning should be displayed.	When clicked place order button without filling payment details, warning displayed.	pass	
4	Whether order table updated after placing order.	Tbl_order must be filled with ordered product details and amount	Tbl_order filled with essential order details.	pass	
5	Whether the total amount is displayed in product page	Total amount should be displayed by calculating with quantity.	Total amount displayed by calculating with quantity	Pass	

6. CONCLUSION

InternHub is a well-structured online platform designed to provide an efficient connection between students, companies, and administrators by integrating essential functionalities for internship management and recruitment. The system enhances accessibility and simplifies the internship search and application process through a centralized and user-friendly environment.

The admin acts as the central authority of the platform, managing company registrations, verifying organizations, monitoring student activities, and supervising internship postings and payment transactions. The admin also ensures system reliability by maintaining data accuracy and overseeing the overall workflow of the platform.

Companies play an important role in providing internship opportunities through the system. After admin approval, companies can post internship programs, manage internship details, review student applications, and approve or reject candidates based on requirements. The platform enables companies to efficiently handle internship recruitment and monitor enrolled students.

Students benefit from an easy-to-use interface that allows them to search verified companies, explore available internships, view detailed information such as duration, syllabus, and fees, and apply for internships online. The system also enables students to track application status and complete payments securely after approval.

7. REFERENCES

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